Python Crash Course

Králik Balázs

March 23, 2026

Outline

Downloads

Optionals are not necessary but really nice to have, I will talk a lot about those.

- Download Python 3.x from `https://www.python.org/downloads/`. Click "Add Python to PATH."
- Optional:ownload Visual Studio Code from `https://code.visualstudio.com/download`
- Optional: Download Git from `https://git-scm.com/downloads`. Make sure to click "Git Bash" option
- Optional:Create a GitHub account at `https://github.com`.
- Optional: For Latex
    - Download MikTex from `https://miktex.org/download`
    - Download Strawberry perl from `https://strawberryperl.com/`

Setup

- Expand the provided PythonCourse.zip file to a location of your choice.
- Cmd prompt at the PythonCourse folder (right click on folder, select "Open with cmd"). For macOS, open terminal and `cd` to the PythonCourse folder.
- type: "`python --version <Enter>`" (or try `python3 --version <Enter>` on macOS), you should see the Python version you installed. If you don't, ask AI until you do (reinstall python for example).
- type: "`python setup_env.py <Enter>`"
- At the end of the script, it says "Next steps: Activate the environment" and tells you what to type. Type that.
- type: "`jupyter notebook <Enter>`". This should open a browser window with the Jupyter interface. This is the basic setup, we can work from here, but I am showing you a better way too.

Better Way

- Open VSCode
- Open folder: **File** → **Open Folder...** and select the unzipped PythonCourse folder.
- On the left side you find the file explorer. Click `python_intro_stats_notebook.ipynb`
- If VS Code asks to install recommended extensions, click **Yes**.
- If VS Code asks what kernel to use, always choose "python virtual environment" and click the one called "venv".
- If VS Code complains about "could not find python interpreter for jupytext", just click away the popup.

Work in Notebooks

- `python_intro_stats_notebook.ipynb`
- `numpy_pandas_guided_tour_finance.ipynb`

Sample Analysis Results

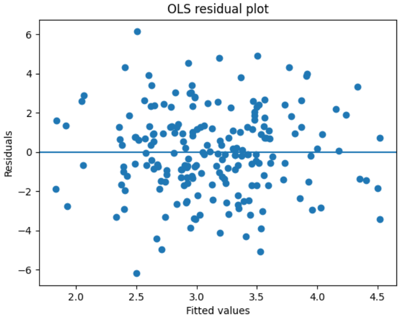

Figure: Some Fancy Analysis

As we can see in Figure 1, our publication deserves a lot of citations.

Collaborating on LaTeX via GitHub

- **Initialize repository**: Create a repo on GitHub, clone it locally
- **Create a branch** for your feature: `git checkout -b feature-name`
- **Edit your .tex file(s)** and commit regularly:
  - `git add <file>`
  - `git commit -m "Clear message about changes"`
  - `git push origin feature-name`
- **Open a Pull Request (PR)** on GitHub for review
- **Resolve conflicts** if others edited the same lines
- **Merge** after approval

Tips for LaTeX Collaboration

- **One sentence per line** in your .tex files—makes diffs clearer and conflicts easier to resolve
- **Commit frequently**—small, logical changes are easier to review and merge
- **.gitignore**—exclude build artifacts:
  - `*.aux`, `*.log`, `*.pdf`, `*.fdb_latexmk`, `*.fls`, `*.nav`, `*.snm`, `*.toc`
- **Communicate**—discuss large structural changes in issues before editing
- **Use descriptive branch names**: `fix-typos`, `add-appendix`, `reformat-intro`

Questions?